Independent Audit of `phase2/uniform_limits.tex` and `phase2/scripts/uniform_limits_*.py`

Independent Audit

November 2026

Abstract

This document reports an independent mathematical and numerical audit of the package `phase2/uniform_limits.tex` (and its compiled `.pdf`) together with its five verification scripts `uniform_limits_analytic.py`, `uniform_limits_scan.py`, `uniform_limits_wh.py`, `uniform_limits_matched.py`, and `uniform_limits_refit.py`. All analytic identities have been re-derived from first principles and all numerical claims have been re-tested with independent code paths in `phase2/scripts/audit_uniform_limits.py` (symbolic series reversion and integrand expansion in SymPy, de Hoog–Knight–Stokes Fourier–Padé Laplace inversion with exact dressed-measure quadrature for the half-line Wiener–Hopf convolution, a Fejér first-rule Chebyshev Nyström solver for the finite-sea Bethe equations, and independent sensitivity refits of all saved datasets). Every package script and `audit_uniform_limits.py` exit cleanly and print `ALL CHECKS PASSED`.

1 Executive Summary and Verdict Table

The package `phase2/uniform_limits.tex` is a disciplined post-errata re-assessment of the semiclassical ($h \to 0$) and edge ($Y = b\kappa^2/h$) limits on the Toda / bad-Boussinesq side. Unlike the earlier Phase 2d–2i packages corrected in `phase2/audit_phase2i.tex` and `phase2/errata.tex`, `uniform_limits.tex` does *not* over-claim a finite-sea resolvent theorem or a settled finite-sea $1/Y$ two-body phase coefficient. Instead, it separates unconditional analytic identities from conditional half-line Wiener–Hopf statements and open finite-sea conjectures **(d)**.

Our independent audit confirms eight of the twelve main claims of the package, qualifies four claims where script coverage or table summaries require explicit caveats, and finds no refuted mathematical formulas in `uniform_limits.tex` itself (while confirming the package's own refutations of the earlier Phase 2d step bound and Phase 2f sign-definite $1/Y$ claim).

Claim in `uniform_limits.tex`	Package label	Audit verdict	Independent audit evidence (`audit_uniform_limits.py`)
1. Bad-Boussinesq / Toda dictionary and fixed-P vs fixed-Y scale separation (§1, Eq. (1))	**(c)**	**Confirmed**	Re-derived $P_0'(\kappa) = 4b\kappa\sinh\kappa > 0$, $v_0 = b\sinh\kappa/\kappa$, $\kappa = \sqrt{3}\beta$ cubic match, and $P_0\|_Y = \frac{4}{3}b^{-1/2}(hY)^{3/2} + O(h^{5/2}) \to 0$. Defect $s = \sigma_{\rm vac}$ vs $\eta\sigma_{\rm vac}$ convention checked.
2. Bulk digamma kernel decomposition, primitive, and Lipschitz operator remainder (§2, Eqs. (2)–(8))	**(c)/(d)**	**Qualified**	Formula and $O(h^2\log(1/h))$ bound confirmed, including exact closed-form cusp constant $2C_{\rm cusp}h^2$ and direct test on an interior Lipschitz function. Qualified only because `uniform_limits_analytic.py` did not directly evaluate the h-dependent cusp integral or test $\mathcal{A}_h$ on a Lipschitz profile.

Claim in `uniform_limits.tex`	Package label	Audit verdict	Independent audit evidence (`audit_uniform_limits.py`)
3. Corrected half-line Wiener–Hopf transform $\mathcal{L}F = \sqrt{\pi}g(s)/s^2$ and inverse series $u(y)$ (§3, Eqs. (9)–(10))	**(c)** cond. / **(d)**	**Confirmed**	Re-derived $g(i\omega)g(-i\omega) = 1/(1-e^{-\lvert\omega\rvert})$, forward $y(u)$ with $-\frac{1}{16\pi^3}(\log u)^2/u^2$, and reversion $u(y)$ through $O(y^{-3})$, confirming A, c, d, e and absence of $\log^k y/y^m$ for $m = 1, 2, 3$.
4. One-body $C_E(y), C_v(y)$ series and phonon-first ($y \downarrow 0$) vs soliton-edge ($y \to \infty$) limits (§3, Eqs. (11)–(12))	**(c)** cond.	**Qualified**	Confirmed $C_E(0^+) = C_v(0^+) = \frac{3}{4\pi}$ vs $\frac{1}{2\pi}$ at $y \to \infty$ (difference $\frac{1}{4\pi}$). Qualified because `uniform_limits_analytic.py` checked the $y \downarrow 0$ limit symbolically only for $C_v(y)$, omitting the small-u expansion of $3J(u)/(4F(u))$ needed for $C_E(0^+)$.
5. Proposition 1: matched half-line two-body expansion $R(Y, r) = c_0(r) + c_1(r)/Y + O(Y^{-3/2}\log^M Y)$ (§4, Eqs. (13)–(19))	**(c)** cond. / **(d)**	**Qualified**	Confirmed $c_0(r)$, $c_1(r)$, and $O(Y^{-3/2}\log^M Y)$ split under stated hypotheses. Qualified because `uniform_limits_matched.py` hard-coded $I_0(t)$ and $I_1(t)$ rather than deriving them symbolically from $8Y^{3/2}t^2 f(u(Y) - u(r^2Y) + u(Yt^2))$; our audit script supplies the missing symbolic derivation.
6. Non-zero $(\log Y)^2/Y$ Taylor term in $\Phi_{\mathrm{cl}}(u(Y), u(r^2Y))$ and cancellation (§4, Eq. (20))	**(c)**	**Confirmed**	Confirmed quadratic Hessian term $\frac{(d_2 - r^2 d_1)^2}{r(1-r^2)Y} = \frac{1-r^2}{4\pi^2 r}\frac{(\log Y)^2}{Y}$ and its cancellation against the dressed-measure outer integral.
7. Endpoint moment μ, $K = -2\mu = 0$, and truncated tail $\mu_{[0,Y]} = -2e/\sqrt{Y} + \cdots$ (§4, Eq. (21))	**(c)** cond. / **(d)**	**Confirmed**	Confirmed $K = -2\mu = 0$ from $\mathcal{L}J = \sqrt{\pi}g(s)/s^3$ having only half-integer powers of y, and confirmed $-2e/\sqrt{Y}$ tail.
8. Renewed read-only one-body fit $C_T = 0.15915494136 \pm 8.23 \times 10^{-6}$ on `onebody_prec_scan.json` (§5)	**(c)**	**Confirmed**	Confirmed non-circular free-intercept fit. Leaving the $1/y$ slope free as well on the 7-point quartic $h \to 0$ values gives $0.1591551100 \pm 5.07 \times 10^{-7}$ (y-spread 5.07×10^{-7} vs 4.73×10^{-7} with fixed $-3d/(2y)$), still bracketing $1/(2\pi)$ within 1.7×10^{-7}.
9. Reproduction of three Phase 2d step-bound violations > 0.004 and linear-h ratio 0.500415 (§5)	**(c)**	**Confirmed**	Independently verified all three violations in `toda_edge_twoterm.json` (0.006326, 0.009016, 0.004512) and ratio 0.500415.

Claim in `uniform_limits.tex`	Package label	Audit verdict	Independent audit evidence (`audit_uniform_limits.py`)
10. Classical-subtracted and unsubtracted Phase 2f refits showing c_1 sign reversal (§5)	**(c)**	**Confirmed**	Independently verified that c_1 changes sign across (h, Y) windows and bases both before and after subtracting Φ_{lat}/h.
11. Endpoint-resolved half-line WH convolution $Q(u(Y), u(r^2Y))$ and negative c_1 fits (§5)	**(c)** cond.	**Confirmed**	Recomputed via independent de Hoog Fourier–Padé inversion and exact dressed-measure quadrature $z = u(w^2)$ (agreement $< 5 \times 10^{-8}$); $c_1 < 0$ across all bases and windows, bracketing formal $c_1(r)$.
12. Fresh fixed-P (κ_1 fixed) and growing-Y finite-sea scans and error budgets (§5)	**(c)**/**(d)**	**Qualified**	Confirmed finite-sea values with independent Fejér Chebyshev Nyström solver ($< 2 \times 10^{-7}$). Qualified by two points: (i) at $\kappa_1 = 0.12$, $1/y = h/(b\kappa_1^2) \in [0.286, 2.29]$ is outside the large-y regime, explaining the h-fit blow-up; (ii) the $\kappa_1 = 0.35$ table row prints the phase mesh diagnostic 4.54×10^{-9} rather than $\max(\text{energy}, \text{phase}) = 4.99 \times 10^{-9}$.

2 Mandatory Six-Point Audit Checklist

1. **Verification of every (c) claim (`audit_uniform_limits.py`).** All **(c)** claims in `uniform_limits.tex` are backed by executable checks, with three script-coverage gaps identified and closed by our independent script:

   - In §2, `uniform_limits_analytic.py` checked the cusp integral $\int_{-1}^{1} R(z/h)|z|\,dz = \frac{h^2}{3}\log(1/h) + O(h^2)$ only indirectly via the moment increment $\int_{1000}^{2000} 2xR(x)\,dx \approx \frac{1}{3}\log 2$, and did not numerically test the operator remainder on a Lipschitz function. In `audit_uniform_limits.py` we evaluate the exact h-dependent cusp integral against its three-term asymptotic series $\frac{h^2}{3}\log(1/h) + 2C_{\mathrm{cusp}}h^2 - \frac{h^4}{60} + O(h^6)$ and test $\mathcal{A}_h f(x_0)$ directly on $f(z) = |z - x_0| + \frac{1}{2}\cos z$.
   - In §3, `uniform_limits_analytic.py` checked the $y \downarrow 0$ limit symbolically only for $C_v(y)$, not $C_E(y)$. In `audit_uniform_limits.py` we derive the $s \to \infty$ expansion $g(s) = 1 + \frac{\pi}{6s} + \frac{\pi^2}{72s^2} + O(s^{-3})$, obtain $F(u) = \sqrt{\pi}u + \frac{\pi^{3/2}}{12}u^2 + O(u^3)$ and $J(u) = \frac{\sqrt{\pi}}{2}u^2 + \frac{\pi^{3/2}}{36}u^3 + O(u^4)$, and verify symbolically that both $C_E(0^+)$ and $C_v(0^+)$ equal $\frac{3}{4\pi}$.
   - In §4, `uniform_limits_matched.py` typed the outer integrands $I_0(t)$ and $I_1(t)$ in Eqs. (16)–(17) by hand before integrating them. In `audit_uniform_limits.py` we derive $I_0(t)$ and $I_1(t)$ symbolically by expanding $8Y^{3/2}t^2 f(u(Y) - u(r^2Y) + u(Yt^2))$ in powers of Y^{-1}.

2. **Honesty of error budgets.** The package separates mesh-refinement diagnostics ($N \to 2N$), $h \to 0$ extrapolation spread across multiple polynomial and logarithmic bases and window lengths, and y- or Y-basis/window spread, adding the three components linearly. Our independent refits confirm that no target value ($1/(2\pi)$ or $c_1(r)$) is fed into any fit.

Two qualifications on the error budgets are recorded in §3: (a) the one-body fit subtracts the analytic half-line tail $-3d/(2y)$ before fitting $(1, y^{-2}, \dots)$, which narrows the y-window spread by a factor of 23 compared with a fully unconstrained $(1, y^{-1}, y^{-2}, \dots)$ fit; and (b) in the fixed-κ_1 table at $\kappa_1 = 0.35$, the "Max. mesh diagnostic" column prints the two-body phase mesh difference 4.54×10^{-9} rather than the one-body energy mesh difference 4.99×10^{-9}.

3. **Conventions: good/bad dictionary and integrable defect s parameter.** The project's canonical equation is the *good* Boussinesq equation $u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}$ with $u = \phi_x$, $L = \frac{1}{2}\phi_t^2 - \frac{1}{2}\phi_x^2 + \phi_x^3 - \frac{1}{2}\phi_{xx}^2$, $\Omega = \int d\pi \wedge d\phi$, one-soliton $k = \sin\alpha$, $v = \varepsilon\cos\alpha$, $M = 2k$, and $P = \frac{2}{3}\varepsilon k^3 \cos\alpha$ **(a)**. The analytic continuation $\alpha = i\beta$ gives the *bad* equation $u_{tt} = u_{xx} + u_{xxxx} + 3(u^2)_{xx}$ with $k = \sinh\beta$, $v = \varepsilon\cosh\beta$, and $P_{\text{bad}} = \frac{2}{3}\varepsilon\sinh^3\beta\cosh\beta$ **(c)**. The package `uniform_limits.tex` works exclusively in the Toda / bad-Boussinesq sector with $a = 1$, $b = e^{-1/2}$, $h = \hbar_T = 3b\hbar_B$, $P_0(\kappa) = 4b(\kappa\cosh\kappa - \sinh\kappa)$, $E_0(\kappa) = 2b^2(\sinh\kappa\cosh\kappa - \kappa)$, and $\kappa = \sqrt{3}\beta$ at leading cubic order $P_B = P_0/(6\sqrt{3}b) = \frac{2}{3}\beta^3 + O(\beta^5)$ **(c)**. Regarding the integrable-defect convention audited across Phase 2 (where scattering transmission uses $s = \sigma_{\text{vac}}$ whereas the bound-state spectrum uses $c = \eta\sigma_{\text{vac}}$), `uniform_limits.tex` does not invoke defect formulas and never conflates s with $\eta\sigma_{\text{vac}}$ **(c)**.

4. **Absence of circularity.** We inspected all five scripts for self-comparisons, target injection, or retroactive tolerance loosening. In `uniform_limits_refit.py`, the intercept y^0 (or h^0, Y^0) and slope Y^{-1} are solved by unconstrained high-precision least squares (`mpmath.qr_solve`). In `uniform_limits_wh.py`, $Q(u(Y), u(r^2Y))$ is computed from numerical contour inversion without using $c_0(r)$ or $c_1(r)$. No circularity is present **(c)**.

5. **Consistency with earlier audits (`audit_phase2i.tex`, `errata.tex`).** The package incorporates every correction mandated by `audit_phase2i.tex` and `errata.tex`: it uses the corrected Laplace transform $\mathcal{L}F(s) = \sqrt{\pi}g(s)/s^2$ (fixing the Phase 2g missing $1/s^2$), retires the invalid $K_0 = 0$ and $O(Y^{-2})$ proofs in favor of a conditional $O(Y^{-3/2}\log^M Y)$ matched bound, reproduces the three Phase 2d > 0.004 step violations, and demonstrates that the Phase 2f finite-sea $1/Y$ coefficient remains sign-indeterminate after full classical lattice subtraction **(c)**.

6. **Foreground script execution and runtimes.** All five package scripts and our independent audit script were executed in the foreground from the repository root and printed `ALL CHECKS PASSED`:

Script	Foreground runtime	Status
`phase2/scripts/uniform_limits_analytic.py`	32.1 s	`ALL CHECKS I`
`phase2/scripts/uniform_limits_scan.py`	1.5 s (cache check; 1026.4 s full scan)	`ALL CHECKS I`
`phase2/scripts/uniform_limits_wh.py`	0.8 s (cache check; 529.2 s full WH)	`ALL CHECKS I`
`phase2/scripts/uniform_limits_matched.py`	20.2 s	`ALL CHECKS I`
`phase2/scripts/uniform_limits_refit.py`	6.1 s	`ALL CHECKS I`
`phase2/scripts/audit_uniform_limits.py`	81.8 s (independent audit)	`ALL CHECKS I`

3 Detailed Claim-by-Claim Audit

3.1 Section 1: Conventions and Fixed-P vs Fixed-Y Limits

Claim 1 ((c), Confirmed). In `audit_uniform_limits.py` (`audit_conventions_and_scales`), we symbolically verified $P_0'(\kappa) = 4b\kappa\sinh\kappa > 0$ for $\kappa > 0$, $dE_0/dP_0 = b\sinh\kappa/\kappa$, and the small-$\kappa$

expansions

$$P_0(\kappa) = \frac{4b}{3}\kappa^3 + \frac{2b}{15}\kappa^5 + \frac{b}{210}\kappa^7 + O(\kappa^9),$$
$$E_0(\kappa) - bP_0(\kappa) = \frac{2b^2}{15}\kappa^5 + \frac{13b^2}{630}\kappa^7 + O(\kappa^9). \tag{1}$$

Setting $\kappa = \sqrt{hY/b}$ at fixed Y yields $P_0 = \frac{4}{3}b^{-1/2}(hY)^{3/2} + \frac{2}{15}b^{-3/2}(hY)^{5/2} + O(h^{7/2}) \to 0$ as $h \to 0$, confirming that fixed-Y limits shrink the classical momentum whereas fixed-$P_0 > 0$ limits send $Y = b\kappa^2/h \propto h^{-1} \to \infty$ **(c)**.

3.2 Section 2: Bulk Digamma Kernel Boundary Layer

Claim 2 ((c)/(d), Qualified). For $K_h(d) = 2\,\mathrm{Re}\,\psi(1 + id/h) - 2\log(1/h) = 2\log|d| + R(d/h)$ with $R(x) = 2\,\mathrm{Re}\,\psi(1+ix) - 2\log|x|$, our script (`audit_kernel_boundary_layer`) independently verifies:

1. The exact primitive $G(L) = \int_0^L R(x)\,dx = 2\,\mathrm{Im}\log\Gamma(1+iL) - 2L\log L + 2L$ to 10^{-61} against numerical differentiation and quadrature **(c)**.

2. The non-alternating Bernoulli tail (since $(-1)^m B_{2m} = -|B_{2m}| < 0$ for all $m \geq 1$):

$$R(x) = \frac{1}{6x^2} + \frac{1}{60x^4} + \frac{1}{126x^6} + \frac{1}{120x^8} + \frac{1}{66x^{10}} + O(x^{-12}),$$
$$G(L) = \frac{\pi}{2} - \frac{1}{6L} - \frac{1}{180L^3} - \frac{1}{630L^5} - \frac{1}{840L^7} - \frac{1}{594L^9} + O(L^{-11}), \tag{2}$$

   and the small-x expansion $R(x) + 2\log|x| = -2\gamma + 2\zeta(3)x^2 - 2\zeta(5)x^4 + O(x^6)$ **(c)**.

3. **Direct cusp-integral evaluation (closing a script gap in `uniform_limits_analytic.py`):** Changing variables $z = hu$ gives $\mathcal{I}_{\mathrm{cusp}}(h) = \int_{-1}^{1} R(z/h)|z|\,dz = 2h^2\int_0^{1/h} uR(u)\,du$. Splitting at $u = 1$ and integrating (2) yields the exact asymptotic expansion

$$\mathcal{I}_{\mathrm{cusp}}(h) = \frac{h^2}{3}\log\frac{1}{h} + 2C_{\mathrm{cusp}}h^2 - \frac{h^4}{60} + O(h^6), \qquad C_{\mathrm{cusp}} = \int_0^1 uR(u)\,du + \int_1^\infty u\Big(R(u) - \frac{1}{6u^2}\Big)\,du \approx -0.25801 \tag{3}$$

   which matches 60-digit quadrature to 4.2×10^{-15} at $h = 10^{-2}$ and 4.2×10^{-21} at $h = 10^{-3}$ **(c)**.

4. **Direct Lipschitz operator test:** For $f(z) = |z - x_0| + \frac{1}{2}\cos z$ at $x_0 = 0.2$ on $[-1, 1]$, the remainder $\Delta_h = \mathcal{A}_h f(x_0) - \big[-\int_{-1}^{1} 2\log|x_0 - z| f(z)\,dz + \pi h f(x_0)\big]$ satisfies $|\Delta_h|/(h^2\log(1/h)) \in \{0.2073, 0.2367, 0.2571\}$ at $h \in \{0.02, 0.01, 0.005\}$, approaching the theoretical cusp coefficient $1/3$ **(c)**. The package's caveat that this direct-operator estimate does not bound $\mathcal{A}_h^{-1}$ up to the endpoints $z = \pm q$ is confirmed **(d)**.

3.3 Section 3: Corrected Half-Line Wiener–Hopf Series and Order of Limits

Claims 3 and 4 ((c) conditional, Confirmed / Qualified). In `audit_uniform_limits.py` (`audit_wh_reversion_and_matched_proof`), we independently expand $g(s) = s^{-1/2}\Gamma(1 + \frac{s}{2\pi})\exp(-\frac{s}{2\pi}\log\frac{s}{2\pi e})$ as $s \to 0^+$ and apply the Hankel contour identity $\mathcal{L}^{-1}[s^{-\nu}\log^m s](u) = (-1)^m\partial_\nu^m[u^{\nu-1}/\Gamma(\nu)]$ to $\mathcal{L}F(s) = \sqrt{\pi}g(s)/s^2$ and $\mathcal{L}J(s) = \sqrt{\pi}g(s)/s^3$. We confirm:

1. The forward expansion $y(u) = [(3/4)F(u)]^{2/3}$ contains the logarithmic term $-\frac{1}{16\pi^3}\frac{(\log u)^2}{u^2}$ **(c)**.

2. Series reversion of $y(u)$ yields $u(y) = y - A\log y + c + \frac{d}{y} + \frac{e}{y^2} + \frac{f_3}{y^3} + O(y^{-4})$ with $A = \frac{1}{2\pi}$, $c = \frac{1-\log(8\pi)}{2\pi}$, $d = \frac{1}{48} - \frac{1}{4\pi^2}$, $e = \frac{4+6\zeta(3)-\pi^2}{96\pi^3}$, and $f_3 = \frac{3\pi^2-36-24\zeta(3)}{576\pi^4}$, with *all* $\log^k y/y^m$ $(k \geq 1)$ terms canceling identically through $m = 1, 2, 3$ **(c)**.

3. In the $s \to \infty$ ($u \downarrow 0$, $y \downarrow 0$) phonon-first regime, Stirling's series gives $g(s) = 1 + \frac{\pi}{6s} + \frac{\pi^2}{72s^2} + O(s^{-3})$, whence $F(u) = \sqrt{\pi}u + \frac{\pi^{3/2}}{12}u^2 + O(u^3)$, $J(u) = \frac{\sqrt{\pi}}{2}u^2 + \frac{\pi^{3/2}}{36}u^3 + O(u^4)$, and $u(y) = \frac{4}{3\sqrt{\pi}}y^{3/2} - \frac{4}{27\sqrt{\pi}}y^3 + O(y^{9/2})$. Substituting into $C_E(y)$ and $C_v(y)$ gives the small-y expansions

$$C_E(y) = \frac{3}{4\pi} - \frac{3}{10}y + \frac{1}{2\sqrt{\pi}}y^{3/2} + O(y^3), \qquad C_v(y) = \frac{3}{4\pi} - \frac{1}{2}y + \frac{1}{\sqrt{\pi}}y^{3/2} + O(y^3), \quad (4)$$

so $C_E(0^+) = C_v(0^+) = \frac{3}{4\pi}$, differing from the $y \to \infty$ limit $\frac{1}{2\pi}$ by $\frac{1}{4\pi}$ **(c)**. Claim 4 is qualified only because `uniform_limits_analytic.py` checked $C_v(0^+)$ symbolically without deriving the small-y series of $J(u(y))/F(u(y))$ for $C_E(0^+)$.

3.4 Section 4: Conditional Proposition 1 and Endpoint Moment

Claims 5, 6, and 7 ((c) conditional / (d), Qualified / Confirmed). Using the exact measure transformation $z = u(x)$ with $B(z)\,dz = 2\sqrt{x}\,dx$ and $x = Yt^2$ ($dx = 2Yt\,dt$), the half-line convolution is

$$Q(u(Y), u(r^2Y)) = 8Y^{3/2}\int_0^r t^2 f\big(u(Y) - u(r^2Y) + u(Yt^2)\big)\,dt. \quad (5)$$

For $r \in (0,1)$ in a compact set, $\sigma = 1 - r^2 + t^2 \geq 1 - r^2 > 0$ keeps the argument $W = u(Y) - u(r^2Y) + u(Yt^2)$ of order Y across the *entire* interval $t \in [0, r]$, including the endpoint layer $t \in [0, L/\sqrt{Y}]$. Writing $W = u(x_W)$ and solving for the dressed coordinate x_W at fixed $t > 0$ gives

$$x_W = Y\sigma + \Delta_0 + \frac{1}{Y}\left[\frac{A\Delta_0}{\sigma} + d\Big(1 - \frac{1}{r^2} + \frac{1}{t^2} - \frac{1}{\sigma}\Big)\right] + O(Y^{-2}), \qquad \Delta_0 = -A\log\frac{t^2}{r^2\sigma}, \quad (6)$$

where both $\log Y$ and c cancel out of Δ_0. Substituting (6) into $f(u(x_W)) = x_W^{-1/2}[1 - (2d + 3A^2)/x_W^2 + O(x_W^{-3})]$ and expanding $8Y^{3/2}t^2[f(u(x_W)) - (Y\sigma)^{-1/2}]$ in SymPy reproduces $I_0(t)$ and $I_1(t)$ in Eqs. (16)–(17) of `uniform_limits.tex` identically, and 60-digit quadrature on $[0, r]$ matches the closed forms $c_0(r)$ and $c_1(r)$ to $< 10^{-54}$ **(c)**. Splitting at $t = L/\sqrt{Y}$ bounds both the inner contribution ($\int_0^{L/\sqrt{Y}} dt = O(Y^{-1/2})$ multiplied by the $O(Y^{-1})$ outer d-term, and $\int_0^{L/\sqrt{Y}} t^2 \log^m Y\,dt = O(Y^{-3/2}\log^m Y)$) and the outer $O(Y^{-2}t^{-2}\log^m Y)$ tail by $O(Y^{-3/2}\log^M Y)$ **(c)**. Claim 5 is qualified only because `uniform_limits_matched.py` hard-coded $I_0(t)$ and $I_1(t)$ rather than deriving them symbolically from (5)–(6).

We also confirm the quadratic Taylor term $\frac{(d_2 - r^2 d_1)^2}{r(1-r^2)Y} = \frac{1-r^2}{4\pi^2 r}\frac{(\log Y)^2}{Y}$ of $\Phi_{\rm cl}(u(Y), u(r^2Y))$ (Claim 6, **(c)**) and the identity $K = -2\mu = 0$ from $\mathcal{L}J(s) = \sqrt{\pi}g(s)/s^3$ having only half-integer powers of y, which implies $\mu_{[0,Y]} = -2e/\sqrt{Y} + O(Y^{-3/2}\log^m Y)$ (Claim 7, **(c)** conditional).

3.5 Section 5: Independent Solver Cross-Checks and Saved-Data Refits

Claims 8–12 ((c)/(d), Confirmed / Qualified).

1. **One-body refit and free-$1/y$ sensitivity (Claim 8, Confirmed (c)):** We confirm the package's read-only refit of `onebody_prec_scan.json`: when the analytic half-line $1/y$ term $-3d/(2y)$ is subtracted prior to fitting $(1, y^{-2}, y^{-3})$ on $y \geq 4, 8, 16$, the free intercept is $C_T = 0.15915494136 \pm 8.23 \times 10^{-6}$ (mesh 5.68×10^{-8}, h-spread 7.70×10^{-6}, y-spread 4.73×10^{-7}), containing $1/(2\pi) \approx 0.15915494309$ within 1.73×10^{-9} **(c)**. As an

independent audit sensitivity check, if the $1/y$ coefficient is *also* left free on the same windows $(4,32),(8,32),(8,64),(16,64)$ of the 7-point quartic $h \to 0$ extrapolated points using bases $(1, y^{-1}, y^{-2})$ and $(1, y^{-1}, y^{-2}, y^{-3})$, the y-window/basis half-spread increases only slightly from 4.73×10^{-7} to 5.07×10^{-7} with center 0.1591551100 (bracketing $1/(2\pi)$ within 1.67×10^{-7}) **(c)**. (By contrast, using only a 4-point quadratic $h \to 0$ fit up to $y = 64$ introduces a -2.6×10^{-5} h-truncation bias because $hy = 0.64$ at $h = 0.01$, showing that the 7-point higher-degree h extrapolation is essential.)

2. **Phase 2d step violations and Phase 2f sign reversal (Claims 9–10, Confirmed (c)):** In `audit_uniform_limits.py`, we independently scan `toda_edge_twoterm.json` and confirm the three step violations > 0.004: $(y_1, y_2) = (3,1)$ from $h = 0.01 \to 0.005$ (0.006326022), $(6,1)$ from $0.01 \to 0.005$ (0.009016413), and $(6,1)$ from $0.005 \to 0.0025$ (0.004511945), with step ratio 0.500415 demonstrating a dominant linear-h error **(c)**. On `matching_toda_data.json` plus `matching_toda_fine.json` ($h \in \{0.04, 0.02, 0.01, 0.005, 0.0025\}$), subtracting Φ_{lat}/h leaves a fitted $1/Y$ coefficient spanning $[-0.050095, +0.028953]$ ($r = 0.4$), $[-0.100731, +0.042933]$ ($r = 0.5$), and $[-0.178826, +0.054938]$ ($r = 0.6$), confirming that the sign of c_1 is not determined by the saved finite-sea grids **(c)**.

3. **Independent de Hoog / dressed-measure WH convolution (Claim 11, Confirmed (c) cond.):** Without calling `uniform_limits_wh.py`, we used de Hoog–Knight–Stokes Fourier–Padé Laplace inversion in $\operatorname{Re} s > 0$ and evaluated $Q(u(Y), u(r^2 Y))$ via the exact dressed-measure integral $8 \int_0^{r\sqrt{Y}} w^2 f(u(Y) - u(r^2 Y) + u(w^2))\, dw$ (which eliminates the $z^{-1/2}$ endpoint singularity of $B(z)$ and requires only large-argument evaluations of f). At $Y \in \{16, 64\}$, $r = 0.5$, our independent residuals $R(16, 0.5) = -0.0208545987$ and $R(64, 0.5) = -0.0198503876$ agree with `uniform_limits_wh.json` to 1.02×10^{-9} and 1.07×10^{-8}, respectively **(c)**. Refitting across 12 model variants (including a $(1, Y^{-1}, Y^{-3/2} \log Y, Y^{-2})$ audit basis) gives strictly negative c_1 in every case ($[-0.007985, -0.007660]$ at $r = 0.4$, $[-0.021454, -0.021442]$ at $r = 0.5$, $[-0.048516, -0.048128]$ at $r = 0.6$), bracketing the formal values $c_1(r) \in \{-0.007737, -0.021445, -0.048436\}$ **(c)**.

4. **Independent Fejér Chebyshev finite-sea solver and fresh-scan audit (Claim 12, Qualified (c)/(d)):** We implemented an independent Nyström solver (`FejerTodaSea`) using Fejér's first rule at Chebyshev roots $x_j = -\cos((2j-1)\pi/(2N))$ on $[-q, q]$. At $h = 0.02$, $\kappa_1 = 0.35$, Fejér quadrature with $N = 360$ gives $q = 1.2417850574$ and $C_E = 0.1594792365$, agreeing with the Gauss–Legendre cache in `uniform_limits_scan.json` ($q = 1.2417849705$, $C_E = 0.1594791631$) to 8.7×10^{-8} and 7.3×10^{-8}, respectively **(c)**. We qualify Claim 12 with two audit observations:

   - *Quantitative cause of small-κ_1 h-fit spread:* At fixed κ_1, the edge variable is $y = b\kappa_1^2/h$, so the large-y expansion parameter is $1/y = h/(b\kappa_1^2)$. At $\kappa_1 = 0.12$, as h decreases from 0.02 to 0.0025, $1/y$ runs from $2.290 > 1$ to 0.286 ($y \in [0.437, 3.494]$). Thus the $h \in [0.0025, 0.02]$ scan at $\kappa_1 = 0.12, 0.18$ sits inside the $y = O(1)$ crossover between $C_E(0^+) = \frac{3}{4\pi} \approx 0.23873$ and $C_E(\infty) = \frac{1}{2\pi} \approx 0.15915$, rather than in the asymptotic $h \to 0$ power-series regime **(c)**.
   - *Minor table discrepancy at $\kappa_1 = 0.35$:* In the fixed-κ_1 table of `uniform_limits.tex`, the "Max. mesh diagnostic" column reports max(energy mesh, phase mesh) for $\kappa_1 \in \{0.12, 0.18, 0.25\}$, but at $\kappa_1 = 0.35$ it prints 4.54×10^{-9} (the two-body phase mesh diagnostic 4.536×10^{-9}) instead of the larger one-body energy mesh diagnostic 4.990×10^{-9} recorded in `uniform_limits_refit.json`. This 4.5×10^{-10} difference is four orders of magnitude smaller than the h-extrapolation spread (5.71×10^{-5}) and does not affect any conclusion **(c)**.

4 Failures Recorded

In accordance with `AGENTS.md`, we record both the package/historical discrepancies asserted by the audit script and the intermediate check failures encountered and fixed while developing `audit_uniform_limits.py`:

1. **Historical Phase 2d step-bound failure (asserted in `audit_uniform_limits.py`):** Three steps in `toda_edge_twoterm.json` exceed the claimed 0.004 uniform bound (0.006326, 0.009016, 0.004512), with linear-h step ratio 0.500415.

2. **Historical Phase 2f $1/Y$ sign-definiteness failure (asserted in `audit_uniform_limits.py`):** Across (h, Y) windows and extrapolation bases on the saved finite-sea data, the fitted $1/Y$ coefficient changes sign both before and after subtracting Φ_{lat}/h.

3. **Audit script development failure 1 (fixed):** In `toda_E0_minus_bP0_series`, our initial hand-calculated κ^7 coefficient of $E_0(\kappa) - bP_0(\kappa)$ was typed as $\frac{1}{42}b^2\kappa^7$ instead of $\left(\frac{8}{315} - \frac{1}{210}\right)b^2\kappa^7 = \frac{13}{630}b^2\kappa^7$. SymPy caught the mismatch immediately; corrected to $\frac{13}{630}$.

4. **Audit script development failure 2 (fixed):** In `fixed_Y_shrinking_momentum`, `sp.series(..., h, 0, 3).removeO()` retained both $O(h^{3/2})$ and $O(h^{5/2})$ because $5/2 < 3$, whereas our initial assertion subtracted only the leading $\frac{4}{3}b^{-1/2}(hY)^{3/2}$ term. Updated to verify both the $h^{3/2}$ and $\frac{2}{15}b^{-3/2}(hY)^{5/2}$ terms.

5. **Audit script development failure 3 (fixed):** In `kernel_G_asymptotic_L100` and `kernel_R_asymptotic_L100`, our initial asymptotic test used alternating signs at $L^{-4}, L^{-6}, \ldots$, forgetting that $i^{2m} = (-1)^m$ combines with $B_{2m} = (-1)^{m-1}|B_{2m}|$ to make every term in $R(L) = \sum_{m\geq 1} \frac{|B_{2m}|}{mL^{2m}}$ positive (and every subleading term in $G(L)$ negative), and required the $m = 5$ term $\frac{1}{66L^{10}}$ at $L = 100$ to pass the strict 10^{-22} threshold without weakening the tolerance.

6. **Audit script development failure 4 (fixed):** In `cusp_integral_asymptotic`, evaluating C_{cusp} with `mp.quad(..., [1, 4, 20, mp.inf])` suffered catastrophic cancellation in $R(u) = 2\,\mathrm{Re}\,\psi(1 + iu) - 2\log u$ at the transformed infinity nodes $u > 10^{30}$ (giving a 2.47×10^{-3} error at $h = 10^{-2}$). Splitting at $u_0 = 10$ and summing the convergent Bernoulli tail $\sum_{m\geq 2} \frac{|B_{2m}|}{m(2m-2)u_0^{2m-2}}$ restored 40-digit accuracy.

7. **Audit script development failure 5 (fixed):** In `audit_wh_reversion_and_matched_proof`, differentiating $\Gamma(\nu)^{-1}$ symbolically left unexpanded half-integer polygammas $\psi^{(r)}(\frac{5}{2} - k)$ and unsplit $\log(8\pi)$ vs $3\log 2 + \log\pi$, and our initial $J(u(y))$ formula omitted the $-\frac{8A}{9}y^{3/2}$ term and factor of 2 in $-\frac{8d}{3}y^{1/2} + \frac{16e}{3}y^{-1/2}$ (since $\frac{16}{9}\cdot(-\frac{3d}{2}) = -\frac{8d}{3}$). Implementing the exact half-integer polygamma recurrence from $\nu = \frac{1}{2}$ and the $\frac{16}{9}$ prefactor resolved both checks identically.

8. **Audit script development failure 6 (fixed):** In `audit_independent_wh_convolution`, our first attempt used a fixed Talbot contour wrapping into $\mathrm{Re}\,s < -2\pi$ and a factor of $4w^2$ instead of $2\,dF = 8w^2\,dw$. Because $\Gamma(1+s/(2\pi))$ has poles at $s = -2\pi n$ on the negative real axis, the Talbot contour crossed those poles and gave an $O(10)$ error at $Y = 16$. Switching to de Hoog–Knight–Stokes Fourier–Padé inversion along a vertical line in $\mathrm{Re}\,s > 0$ and using $8w^2\,dw$ reproduced $u(Y)$ to $< 10^{-24}$ and $R(Y, r)$ to $< 1.1 \times 10^{-8}$.

9. **Audit script development failure 7 (fixed):** In `FejerTodaSea`, our initial prototype solved $\rho(q) = 0$ in hyperbolic rapidity coordinates rather than the unit-density normalization $\int_{-q}^{q} \rho(x)\,dx = 1$ in bare momentum $x \in [-q, q]$ with bare energy $x^2/2$. Enforcing

$\int_{-q}^{q} \rho(x)\,dx = 1$ on Fejér Chebyshev nodes matched the Gauss–Legendre Fermi momentum and one-body energy to $< 2 \times 10^{-7}$.

References

[1] NIST Digital Library of Mathematical Functions, https://dlmf.nist.gov/, Chapter 5 (Gamma Function), verified for the digamma and log-gamma Stirling series used in §2–§3.